# ETSI TS 125 452 V13.0.0 (2016-01)




**Universal Mobile Telecommunications System (UMTS);**
**UTRAN Iupc interface: signalling transport**
**(3GPP TS 25.452 version 13.0.0 Release 13)**



Reference
RTS/TSGR-0325452vd00

Keywords
UMTS

***ETSI***

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# Foreword

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Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# 1 Scope

The present document specifies the signalling transport related to PCAP signalling to be used across the Iupc interface. The Iupc interface is a logical interface for the interconnection of Stand-Alone SMLC (SAS) and Radio Network Controller (RNC) components of the Universal Terrestrial Radio Access Network (UTRAN) for the UMTS system. The radio network control signalling between these nodes is based upon the Position Calculation Application Part (PCAP).

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TS 25.422: "UTRAN Iur Interface Signalling Transport".

[2] ITU-T Recommendation Q.711 (1996-07): "Functional description of the signalling connection control part".

[3] ITU-T Recommendation Q.712 (1996-07): "Definition and function of Signalling connection control part messages".

[4] ITU-T Recommendation Q.713 (1996-07): "Signalling connection control part formats and codes".

[5] ITU-T Recommendation Q.714 (1996-07): "Signalling connection control part procedures".

[6] ITU-T Recommendation Q.715 (1996-07): "Signalling connection control part user guide".

[7] ITU-T Recommendation Q.716 (1993-03): "Signalling Connection Control Part (SCCP) performance".

[8] 3GPP TS 25.305: "Stage 2 functional specification of UE positioning in UTRAN"

# 3 Definitions and abbreviations

## 3.1 Definitions

For the purposes of the present document, the following definition applies:

**Stand-Alone SMLC (SAS):** As defined in TS 25.305 [8].

## 3.2 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| PCAP | Position Calculation Application Part |
| SAP | Service Access Point |
| SAS | Stand-Alone SMLC |
| SCCP | Signalling Connection Control Part |
| SMLC | Serving Mobile Location Centre |

| | |
|---|---|
| UE | User Equipment |
| UMTS | Universal Mobile Telecommunication System |
| UTRAN | UMTS Terrestrial Radio Access Network |

# 4 PCAP Signalling Bearer

## 4.1 Introduction

This clause specifies the Signalling Bearer protocol stack that supports the PCAP signalling protocol.

The following requirements on the Signalling Bearer can be stated:

- provide reliable transfer of control plane signalling messages in both connectionless mode and connection-oriented mode;
- provide separate independent connections for distinguishing individual transactions;
- provide networking and routing functions;
- provide redundancy in the signalling network;
- provide load sharing.

## 4.2 Signalling Bearer

The Iupc signalling bearer shall comply with the requirements of clause 5.2 in TS 25.422 [1].

## 4.3 Services Provided by the Signalling Bearer

When considering the requirements that the upper layers, i.e. PCAP, have on the Signalling Bearer, there are a number of services it has to provide and a number of functions to perform. These numbers of services that the signalling bearer shall provide, to the upper layers, are stated in references ITU-T Rec. Q.711 [2], ITU-T Rec. Q.712 [3], ITU-T Rec. Q.713 [4], ITU-T Rec. Q.714 [5], ITU-T Rec. Q.715 [6], and ITU-T Rec. Q.716 [7].

# Annex A (informative): Change history

| Date / TSG | TSG Doc. | CR | Rev | Subject/Comment | New |
|---|---|---|---|---|---|
| 12/2008 | - | - | - | Creation of Rel-8 version based on v7.1.0 | 8.0.0 |
| 12/2009 | - | - | - | Creation of Rel-9 version based on v8.0.0 | 9.0.0 |
| SP-49 | SP-100629 | | | Clarification on the use of References (TS 21.801 CR#0030) | 9.0.1 |
| 03/2011 | | | | Creation of Rel-10 version based on v9.0.1 | 10.0.0 |
| 52 | RP-110685 | 0003 | 1 | References cleanup (25.452) | 10.1.0 |
| 09/2012 | | | | Update to Rel-11 version (MCC) | 11.0.0 |
| 09/2014 | | | | Update to Rel-12 version (MCC) | 12.0.0 |
| 12/2015 | | | | Update to Rel-13 version (MCC) | 13.0.0 |

# History

| Document history | | |
|---|---|---|
| V13.0.0 | January 2016 | Publication |
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